th# Theory of evolutionary algorithms

History, (1+1) EA $\approx$ (1+1)ES, Linear Pseudo boolean functions, Methods to study EA convergence: Fitness based partitions, the coupon collector problem, expected multiplicative distance decrease. Geometric operators.

Original Schema Theory

What is a schema?
An ex. individual is a binary string. A schema is a way to say that i am interested only on the firs two bits of that string, or on a subset of them.

A **schema** is a string in $\{0,1,*\}$. Given a schema $s \in \{0,1,*\}^n$ and an individual $x \in \{0,1\}^n$ we say that x contains s when $\forall i \in 1,\dots,n : x_i = s_i$ if $s_i \in \{0,1\}$ ex. 0101 contains $*1*1$. The original theory wants to study what are the schema that survive generations.

The **lenght** of a schema is the difference between the index of the leftmost not "$*$" position and the index of the rightmost not "$*$" position plus one. The length of `0**1*1` is 6 while `**010*` has length 3.

What is the probability of making a schema by performing a one point crossover? The order of f a schema is the number of don't care `*` symbols in it.

$$\left(1-\frac{1}{n}\right)^{\text{order}(s)}$$

Schemata with a small order and small lenght where the average fintess of an individal that contains that schema is high survive more. This is a result that is good only for the first generation.

Other approaches appeared

EA as Markov Processes.

Will the EA converge to a stationary distribution where all populations containing a global optimal have prob 1? Do we always converge to good solutions?

If s_i, s_j is populations then $P(s_i|s_j)$ is the mutation probability then we have mutation we have that this is positive but small.

If M is the transition matrix then $M_{ij} = P(s_i|s_j)$ then M has only positive entries. So the only stationary distribution is the one satisfying $x = Mx$. But all populations will have positive probability. But if we add elitism the process will converge to a distribution where only the populations with the global optimum have positive probabilities. Note that this is also true if i do random sampling while keeping the best solution.

The main way to study EA is

Runtime analysis

Ingredients:

- An evolutionary algorithm A
- A problem or a class of problems P
- What is the expected time for A to find the optimum for any problem the class P.

Study 1+1 EA.

```
// (1 + 1) EA

x <- Random {0,1}^n
forever
{
    offspring <- Mutate(x)
    x <- Fittest(x, offspring)
}
```

Linear pseudo-boolean funtion:
$f : \{0,1\}^n \to \mathbb{R}$ for $x = (x_1, x_2, \dots x_n)$ we have a fitness that is $f(x) = w^T x$ with $w_i \in \mathbb{N}$. The global optimum when maximizing is $\underline{1} = (1,1,\dots,1)$. If all $w_i = 1$ then we have the onemax problem. If $w_i = 2^{i-1}$ we have that $f(x) = \text{binary number rapresented by x}$.
Questions:

1. What is the expected optimization time of the (1+1)EA on onemax?
2. What is the expected optimization time on any linear pseudo-boolean function?

Onemax (and some other considerations)

Fitness based partitions:
Taking the solution space $S = \{0,1\}^n$ take a partition $A_1, \dots, A_m$ of S. We define a partial ordering saying that $A_i <_f A_j$ according so that $\forall a \in A_i,\ \forall b \in A_j :\ f(a) < f(b)$ If

- $A_1 <_f A_2 <_f \dots <_f A_m$ and A_m contains the global otpimum than we say to have $A_1, \dots, A_m$ to be a fitness based partitions. In (1+1)EA you can only go up the partition ladder.

Given $x \in A_i$ an individual $P(x)$ the probability of it to have an offspring in $A_{i+1} \cup A_{i+2} \cdots$ the expected time to jump further is $1/P(x)$. We define:

$$p_i = \min_{a \in A_i} P(a)$$

So $1/p_i$ is tje expected time to *jump over to the better fitness level. Worst case when we jump from only from $A_i \rightarrow A{i+1}. $Shxpimahhpimmisbnabby$:$ \sum{i=1}^{n-1} \frac{1}{p_i} $Bwnfinhispbabiiisp_i$. (for onemax) $A_0 = 0^n$ ... In general a solution $x \in A{n-k}$ if k bits of x are zeros.*

Theorem

The expected optimization time for a (1+1)EA for the onemax problem is $O(n \log n)$.
The probability of moving from A_{n-k} to A_{n-k+1} is $\leq$ than p_{n-k}. As the probability of flipping a bit from 0 to 1 is

$$(k/n)(1-1/n)^{n-1}$$

Prob of flippoing a 0 bit to a 1 bit times the prob of not flipping any other bit. So this is the probability of flipping exactly one 0 bit to a 1 bit. This is

$$(k/n)(1-1/n)^{n-1} > k/(an)$$

So $p_{n-k} \leq k/an$ Implies $1/p_{n-k} \leq an/k$

$$\sum_{k=1}^{n} \frac{1}{p_{n-k}} = an \sum_{k=1}^{n} = O(n \log n)$$

Next: lecture-13-optimization-for-ai